

Power Electronics — Chapter 1

DC-DC Converters: Buck Converter

POWER_E_466

The buck converter is a DC-DC converter that steps down voltage ($V_{out} < V_{in}$).

Motivating the topology

Start from a black-box model: given $V_{in} = 50\text{ V}$, we want an average output of $V_{out} = 20\text{ V}$ across a resistive load.

Ideal switch model

The simplest way to reduce an average voltage is to switch the input on and off periodically (an ideal SPDT switch connecting the load either to V_{in} or to ground), so that the average value is controlled by the fraction of time spent connected to V_{in} — the *duty cycle* D :

$$\langle V \rangle = D V_{in}.$$

Why we need an LC filter

The switched (PWM) waveform is not a clean DC voltage — it still contains switching-frequency ripple. From a Fourier-series point of view, the desired DC output corresponds only to the $f = 0$ Hz (DC offset) term; every other harmonic should be attenuated. A low-pass *LC filter* does exactly this: it passes the DC component while suppressing the switching-frequency content.

Inductor and capacitor averaging identities

For an inductor,

$$v = L \frac{di}{dt} \iff i = \frac{1}{L} \int v dt,$$

so if i is periodic in steady state, the average inductor voltage must be zero:

$$\langle v_L \rangle = 0 \quad (\text{volt-second balance}).$$

For a capacitor,

$$i = C \frac{dv}{dt} \iff v = \frac{1}{C} \int i dt,$$

so if v is periodic in steady state, the average capacitor current must be zero:

$$\langle i_C \rangle = 0 \quad (\text{current-second/charge balance}).$$

Steady-state inductor current ripple

Over one switching period T_s , the inductor current i_L ramps up while the switch connects to V_{in} (duration DT_s) and ramps down while it connects to ground (duration $(1 - D)T_s$):

$$\frac{di_L}{dt} = \frac{v_L}{L}, \quad 2 \Delta i_L = \frac{V}{L} DT_s \quad (\text{rising}), \quad 2 \Delta i_L = \frac{V}{L} (1-D)T_s \quad (\text{falling})$$

Worked example: switch ON

Given $V_{in} = 50\text{ V}$, $D = 0.5$, $L = 10\text{ mH}$, $T_s = 1\text{ ms}$, and (for this simplified example) an output branch modelled as a current source with $V = 0$:

$$\frac{di_L}{dt} = \frac{50}{10^{-3}} = 50 \times 10^3\text{ A/s} \longrightarrow \Delta i_L = 5\text{ A}.$$

At $t = 0^+$: $i_L = 5\text{ A}$, $i_O = 0$, $i_C = i_L - i_O = 5\text{ A}$.

Worked example: switch OFF

$$\frac{di_L}{dt} = \frac{0}{10^{-3}} = 0 \text{ A/s},$$

so i_L stays constant at 5 A while the switch is off (for this particular example, since the inductor sees 0 V in both states here).

Is this steady state?

Not necessarily — V_{out} can still change from cycle to cycle if the circuit is not yet in continuous, periodic operation. As i_O increases from 0, $i_C = i_L - i_O$ decreases; by charge balance, $\langle i_C \rangle = 0$ is exactly the condition that defines the steady-state operating point.

Approaching steady state

Classic multi-period inductor current waveform $i_L(t)$ ramping toward steady state: slope $\frac{V_g - v(t)}{L}$ while on, slope $\frac{-v(t)}{L}$ while off, converging so that $i_C \approx 0$ in steady state (cf. Erickson, *Fundamentals of Power Electronics*).

Realizing the switch with real devices

The ideal SPDT switch (poles A , B , C) is not a physical component. It is realized using one *active* switch (a transistor, controlled by a gate/base signal) for the A – B throw, and one *passive* switch (a diode, which conducts automatically depending on the circuit's own voltage/current) for the C – B throw.

Switch analysis: node A–B (transistor position)

When the ideal switch is ON: $V_{AB} = 0$, $I_{AB} = I_L$. When OFF: $V_{AB} = V_S$, $I_{AB} = 0$. This (I, V) locus (first quadrant only: on along the current axis, off along the voltage axis) matches an *active* switch such as a MOSFET/BJT/IGBT.

Switch analysis: node C-B (diode position)

When the ideal switch is ON (meaning the $A-B$ throw is closed):
 $V_{CB} = -V_S$, $I_{CB} = 0$. When OFF (meaning $C-B$ conducts instead):
 $V_{CB} = 0$, $I_{CB} = I_L$. This locus (blocks negative voltage, conducts positive current, no external control terminal) matches a *passive* switch: a diode.

Switch device characteristics

- **BJT / IGBT** are active switches: the conducting state is set entirely by the control-terminal signal, independent of the waveforms at the power terminals. Ideal characteristic: $i = 0$ when off (blocks $v \geq 0$); $v \approx 0$ when on (conducts $i \geq 0$). Poor/no reverse-conducting or reverse-blocking capability.
- **MOSFET** behaves similarly, but can additionally conduct in reverse. It is normally operated with $i \geq 0$, same as BJT/IGBT, so an active SPST switch can be realized with a BJT, IGBT, or MOSFET.
- **Diode** is a passive switch with no control terminal: its state is determined entirely by the terminal waveforms $i(t)$, $v(t)$. Ideal characteristic: off ($i = 0$) when $v < 0$; on ($v = 0$) when $i > 0$. It can block negative voltage but not positive voltage.

$$\boxed{V_O = D V_S} \quad \text{with } \langle i_C \rangle \rightarrow 0 \text{ in steady state.}$$